PRIYANSH PATEL

AI/ML Engineer Intern — PyTorch — NLP — Computer Vision — RAG

Ahmedabad, Gujarat, India • +91-8780772357 • ppriyansh69@gmail.com

github.com/honeybadger-REX • linkedin.com/in/priyansh-patel-659536275 • huggingface.co/priyanshbadge

SUMMARY

Computer Engineering undergraduate specializing in AI and Data Science with hands-on experience building machine learning, deep learning, NLP, computer vision, Transformer, and retrieval-augmented generation systems. Experienced in developing PyTorch models, RAG pipelines, FastAPI inference services, and Dockerized applications. Seeking an AI/ML engineering internship.

SKILLS

Languages: Python, SQL, C++ (Networking)
ML/DL: PyTorch, Scikit-learn, CNNs, Random Forest, Transformers, Model Evaluation
NLP/GenAI: RAG, CRAG, Self-RAG, Embeddings, Semantic Search, FAISS, Sentence-Transformers, Hugging Face
Computer Vision: OpenCV
Backend/Systems: FastAPI, REST APIs, Docker, MediaMTX, FFmpeg
Tools: Pandas, NumPy, Matplotlib, Jupyter, Google Colab, Git/GitHub, DSA

PROJECTS

English-to-German Transformer from Scratch — PyTorch | GitHub
• Implemented an encoder-decoder Transformer with multi-head self/cross-attention, positional embeddings, padding masks, and causal masks for English-to-German translation.
• Improved test BLEU from 12.5 to 19.0 by expanding training data to 20,000 Multi30k pairs and diagnosing word-choice and data-coverage issues.
• Added masked-token reconstruction to evaluate learned encoder representations.

CRAG & Self-RAG Systems — Python, FAISS, Transformers, FastAPI, Docker | GitHub
• Developed RAG pipelines with document chunking, Sentence-Transformer embeddings, FAISS retrieval, and LLM-based answer generation.
• Implemented CRAG to evaluate retrieved context and filter low-relevance information before generation; built Self-RAG with retrieval, reranking, and answer-quality evaluation components.
• Containerized the application with Docker and exposed the inference workflow through FastAPI for reproducible deployment.

Exoplanet Transit Detection — 1D-CNN, BLS, Random Forest | ISRO Bharatiya Antariksh Hackathon 2026
• Developed a hybrid 1D-CNN, Box Least Squares, and Random Forest pipeline for detecting exoplanet transit signals from astronomical light curves.
• Packaged the inference workflow as a reusable FastAPI service.

AI Video Surveillance Platform — FastAPI, MediaMTX, FFmpeg, Docker | GitHub
• Developed a FastAPI backend with authentication, camera/vendor/event management, SQL database integration, and a JavaScript dashboard.
• Integrated MediaMTX for live streaming and FFmpeg-based RTSP health checks to monitor camera availability; computer-vision inference integration is in progress.

Bone Image Classifier — Deep Learning | Hugging Face
• Built a deep-learning image-classification application for bone images and deployed the trained application on Hugging Face Spaces.

EDUCATION

Silver Oak College of Engineering and Technology, Ahmedabad | Expected May 2027
B.Tech in Computer Engineering — CGPA: **8.06/10**
AI & Data Science Honors, Silver Oak University — 10 credits completed
Coursework: Machine Learning, Deep Learning, Computer Vision, DSA, Probability & Statistics, OOP, DBMS

EXPERIENCE & LEADERSHIP

Technical Competition Coordinator — College Technical Fest
• Designed web-development challenges and coordinated participant registration, submissions, and evaluation.

Participant — Smart India Hackathon
• Prototyped a citizen complaint system using image-similarity based duplicate-complaint detection and an authority notification workflow.

CERTIFICATIONS

Machine Learning — Stanford Online (Coursera), 2026